

Guidelines of the Competition

The competition will feature three stages: the pre-screening round, the semifinal round, and the final round. Details of the format and evaluation criteria for each round are outlined below.

1.0 THEME

“Innovative and Affordable Healthcare Solutions for Local Challenges”

This theme invites participants to design biomedical and healthcare innovations (hardware devices, software applications etc.) that tackle healthcare challenges unique to Sri Lanka, such as non-communicable diseases, rural healthcare accessibility, and public health concerns. The goal is to develop affordable, impactful, and scalable solutions.

Sub-Themes:

1. Empowering Rural Healthcare

Design portable, easy-to-use devices to bridge the urban-rural healthcare gap.

Example: A portable diagnostic kit for chronic kidney disease or dengue.

2. Personalized Health Monitoring

Develop wearable or mobile-based solutions for continuous health monitoring.

Example: A wearable device to monitor vital signs such as heart rate, SpO2, and stress levels.

3. Assistive Technology for Inclusivity

Create devices to enhance the quality of life for individuals with disabilities.

Example: A low-cost, smart prosthetic limb or a communication device for paralyzed patients.

4. Sustainable Biomedical Engineering

Innovate eco-friendly devices for sustainable healthcare practices.

Example: Solar-powered sterilization units or biodegradable medical tools.

1.1 PRODUCT SPECIFICATIONS

- The product must address a healthcare-related issue and should be compatible with modern technology.
- The product can be either software or hardware, and you should be able to prove that it is a product of your own.
- Devices with physical enclosures should have openable designs to allow for

inspection.

- The teams have the freedom to select a contemporary issue related to healthcare and provide a solution. Section 1.7 discusses the nature of these problems and corresponding solutions with examples. However, they are not expected to limit the teams.
- Solutions/products that align with the competition theme are highly encouraged.

1.2 INSTRUCTIONS FOR THE PARTICIPANTS

- Teams can consist of a minimum of 3 and a maximum of 5 participants. All participants should be enrolled in an undergraduate program at a Sri Lankan University. They should provide valid identification documents (university ID or record book) to prove their eligibility.
- A team doesn't need to include participants from a single university. Brainstorm organizers would like to encourage the participants to collaborate with undergraduates from different disciplines (Engineering, Medicine, Biology, Chemistry, Physics, and Computer Science) and universities. A team can have a mentor of their own. (optional)
- Each participant may join only one team.
- A team can submit multiple solutions. However, only one solution will be selected for the final round from one team from the pre-screening round.
- Please note that Brainstorm's past finalists (1, 2, 3 positions) will not be able to register the same project they showcased unless they can prove that they have introduced a significant improvement. The significance of the improvement will be evaluated by the judges.
- Teams must submit a project proposal for the pre-screening round, from which semifinalists will be shortlisted. More information about this can be found in section 1.4.
- In the semifinal round, the teams must submit a video pitching their solution. Rules will be properly provided once the pre-screening round is over.
- In the final round, the evaluation will be based on the technical, economic, and social aspects of the solution and the product. The teams are expected to present their solutions and demonstrate their product/prototype clearly and concisely. More information about the final round and its evaluation process will be provided before the commencement of the semi-final round.
- The teams must be able to prove that it is a product of their own.
- Updates about the competition will be posted on the University of Moratuwa EMBS Student Branch Chapter Facebook page and emailed to the participants.
- A series of workshops will be held before the proposal submission deadline, and all contestants are encouraged to participate. Details about the workshops will be shared on our official Facebook page. The knowledge gained through these

workshops will be evaluated in the semi-final and the final rounds.

1.3 REGISTRATION

The registration link for Brainstorm 2026 will open on March 12th at 6:00 PM.

1.4 PRE-SCREENING ROUND

All teams are requested to submit a project proposal covering,

- The problem that is being addressed.
- The process of information collection about the problem.
- A brief description of the solution, including an overview, methodology, functional diagrams, and required resources.
- Impact of the solution on the healthcare sector.

A sample proposal template can be [found here](#)(link for sample proposal).

1.5 Semifinal round

- Teams selected from the pre-screening round are eligible to participate in this round.
- This round will be conducted online. Each team must submit a video of a given duration, pitching their solution.
- More details will be provided after the results of the pre-screening round are announced.

1.6 Final round

- Teams selected from the semifinal round will participate in the final round.
- Each final team will be assigned a mentor. They will have a month to develop the prototype of the proposed idea before the Grand Finale.
- Please note that a working product/prototype of the solution and a live demonstration are mandatory for this round. Teams that fail to provide one will be disqualified.
- In this round, the evaluation will be based on the following criteria.
 - Technical implementation / Use of suitable technology
 - Innovation, originality, creativity
 - Quality of the idea of the presentation

- Financial Feasibility
- Development cost
- Return on Investment (ROI)
- Affordability to the customer
- Market demand
 - Readiness to market
 - Social Impact
 - Safety and effectiveness
- Regulatory approval process
 - The category the device belongs to
 - Pre-clinical and clinical trials that should be performed

(Knowledge required for this will be covered in workshops before the competition)

- More details will be provided before the commencement of the final round.

1.7 Sample Problems and Design Ideas

Sample Problems

Design ideas from previous years

- 2024 champions - Team Braineocare

AI-Powered User-Friendly Neonatal Seizure Detection Device

Neonatal seizures, which occur within the first four weeks of life, pose significant risks to the developing brain. They are often challenging to detect due to their subtle manifestations, which can be mistaken for normal physiological behaviors. Accurate diagnosis relies on EEG monitoring, which is reliable but resource-intensive, requiring the expertise of neurologists, lengthy preparation, and high costs.

Solution of Team Braineocare is a low-cost, AI-powered device for detecting neonatal seizures. As a preliminary step, the device was tested on children aged 4 to 14 who experience absence seizures, a milder and more common seizure type, to evaluate its performance. The solution involves collecting EEG data during induced absence seizures, comparing the results with standard EEG equipment, and gathering user feedback on comfort and experience. This approach ensures a safe and ethical method for improving diagnostic outcomes for vulnerable neonates.

- 2023 champions - Team FlashGuard

Smart Glasses for Real-Time Protection from Photosensitive Epileptic Triggers

Epilepsy is a chronic brain disorder that affects over 40 million people worldwide. One

specific form of epilepsy is photosensitive epilepsy (PSE), where seizures are triggered by visual stimuli, such as flashing lights or rapidly moving images. PSE is particularly common among younger individuals, and it occurs more frequently in women. The condition can significantly impact daily life, with symptoms ranging from unusual sensations to loss of consciousness and behavioral changes during seizures. Current treatments, including medications, often have limited effectiveness and may cause harmful side effects. Existing preventive measures, such as blue light glasses, seizure-detection devices, and video analysis software, do not provide comprehensive real-time protection against visual triggers.

FlashGuard is a pair of smart glasses designed to detect visual triggers in real time and automatically darken the lenses to protect the wearer. Accompanied by a mobile app for feedback and customization, FlashGuard offers a non-invasive, personalized solution to enhance the quality of life for individuals with PSE.

Contact details

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